CLINICAL · NOSE & SINUSES

Juvenile nasopharyngeal angiofibroma

A teenage boy with a blocked nose and nosebleeds that keep coming back has a juvenile angiofibroma until proven otherwise. It is benign, but it bleeds torrentially, grows through the natural doorways of the skull base, and is diagnosed by imaging, never by a biopsy in clinic.

AT A GLANCE

- Adolescent boy, blockage and heavy bleeding.
- Image, don't biopsy.
- Embolise, then remove.
- Drill the pterygoid base.
- UPMC staging adds residual ICA vascularity after embolisation.

1 CHAPTER ONE Overview

The bleeding tumour of adolescent boys.

1.1 Definition

Juvenile nasopharyngeal angiofibroma (also called nasopharyngeal fibroma) is a benign, locally aggressive, highly vascular tumour arising at the back of the nasal cavity, near the sphenopalatine foramen.

1.2 Epidemiology

The commonest benign tumour of the nasopharynx, yet under 1% of head and neck tumours. Almost exclusively in boys and young men of 10 to 20 years.

1.3 Why it matters

Profuse epistaxis, anaemia, and spread into the orbit and skull base. Biopsy in clinic can cause catastrophic bleeding. It does not regress on its own, so waiting is not an

option.

2 CHAPTER TWO Aetiology and pathology

Hormones and malformed vessels.

2.1 Why adolescent boys

It is thought to be testosterone-dependent: a hamartomatous nidus of vascular tissue, perhaps a remnant of the first branchial arch artery, is activated at puberty. About three-quarters of tumours carry androgen receptors.

2.2 Molecular clues

Vascular endothelial growth factor and insulin-like growth factor II drive growth; stromal cells show beta-catenin activity, and C-MYC is found in large tumours. Angiofibroma is about 24 times commoner in familial adenomatous polyposis of the colon.

2.3 Site of origin

The posterior nasal cavity, at the superior margin of the sphenopalatine foramen, near the junction of the sphenoidal process of the palatine bone with the root of the pterygoid plate.

2.4 Gross and microscopic pathology

A reddish-purple, lobulated, sessile mass covered by mucosa, without a capsule. Two components: endothelium-lined vascular spaces lacking muscle and elastic tissue (so they cannot contract and bleed profusely), and a stroma of stellate and spindle cells laying down collagen, whose amount decides how firm the tumour is.

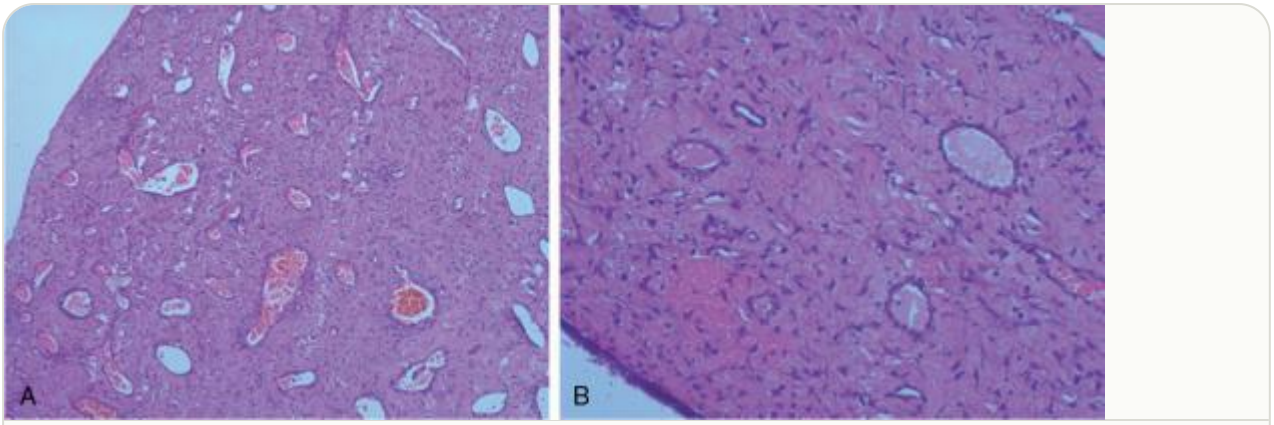


FIGURE 1 Angiofibroma: dilated vascular spaces in a fibrous stroma.
Dhingra, fig. 49.4

3 CHAPTER THREE Spread

Through the doorways of the skull base.

3.1 Three modes

- **Along paths of least resistance:** into the nose and nasopharynx, and through fissures and foramina: the foramen rotundum, vidian (pterygoid) canal, palatovaginal canal and inferior orbital fissure.
- **By expansion:** pushing the posterior wall of the maxillary sinus forwards from the pterygopalatine fossa (the antral sign).
- **By bone destruction:** at the root of the pterygoid and the greater wing of the sphenoid.

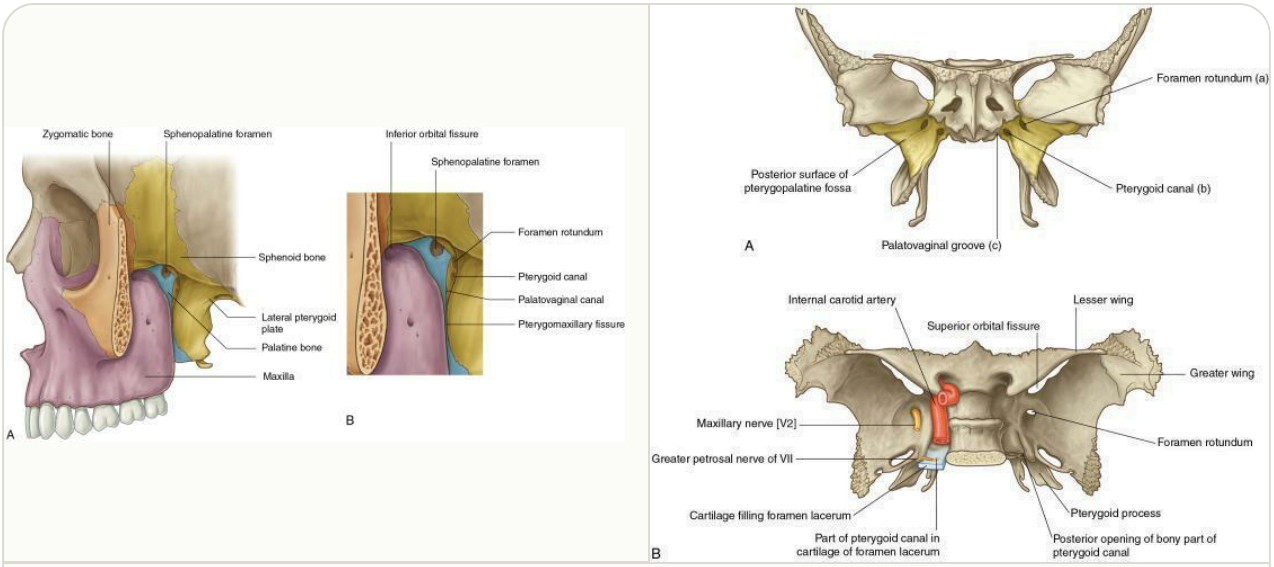


FIGURE 2 First, the pterygopalatine fossa from the side, with its routes to the orbit, infratemporal fossa and cranial cavity. Second, the openings in its posterior wall: foramen rotundum, vidian canal and palatovaginal canal.

Dhingra, figs 49.1 and 49.2

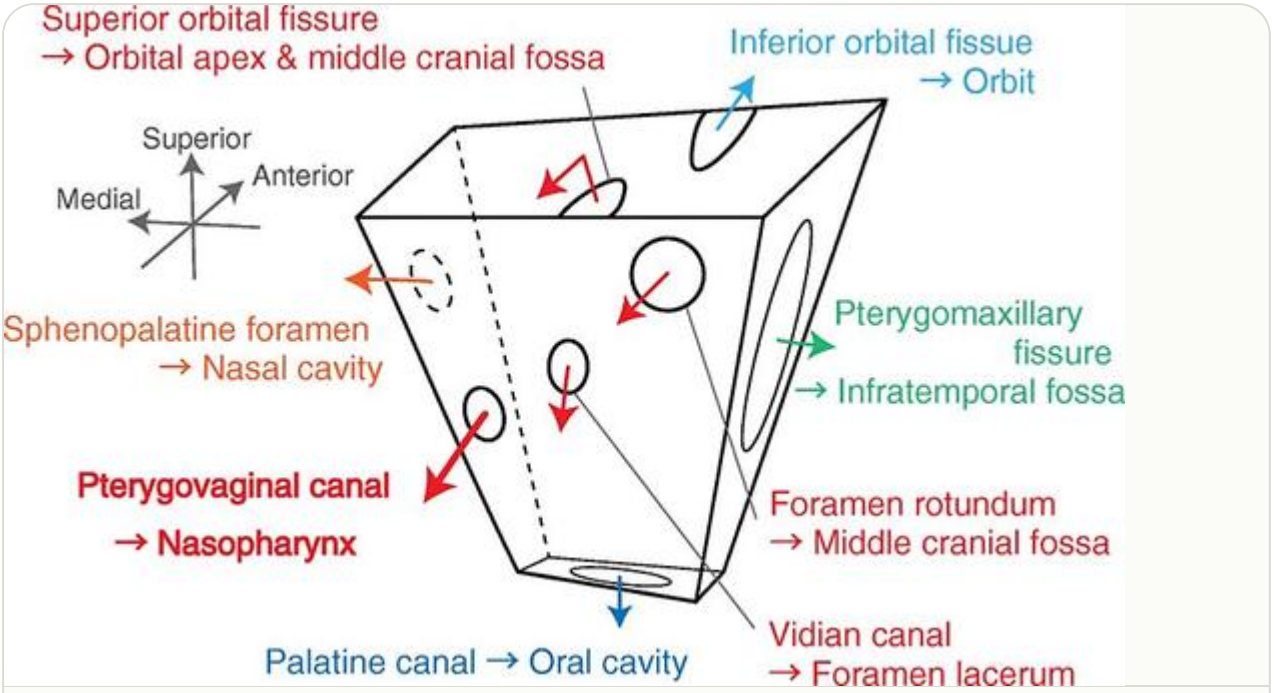


FIGURE 3 Communications of the pterygopalatine fossa, the crossroads through which the tumour spreads.

Anatomy atlas · ENT Steps p. 161

3.2 Extensions

1. Nasal cavity and nasopharynx, blocking the choanae and displacing the septum.
2. Paranasal sinuses: maxillary, sphenoid, ethmoid.

3. Pterygopalatine fossa, then through the pterygomaxillary fissure into the infratemporal and temporal fossae and the cheek.
4. Orbit, through the inferior orbital fissure: proptosis, diplopia, optic nerve compression.
5. Cranial cavity in 20 to 25%, mainly the middle fossa (lateral to the internal carotid if it erodes the floor, medial if it comes through the sphenoid and sella), less often the anterior fossa; usually extradural, sometimes into the cavernous sinus.

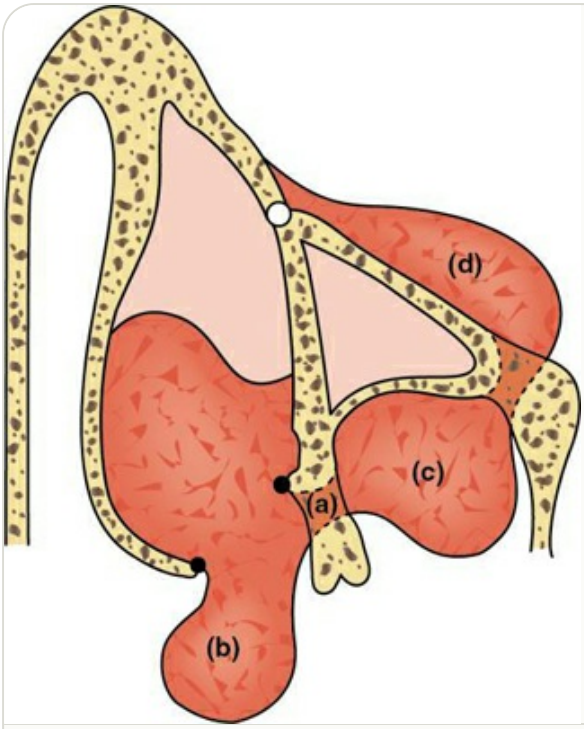


FIGURE 4 Axial view of the spread: into the nose, the nasopharynx, behind the antrum into the pterygopalatine fossa, and on into the cheek.

Dhingra, fig. 49.3

RECURRENCE HIDES IN CANALS

Tumour left in the vidian canal or the basisphenoid is the commonest source of recurrence: drill the canal at surgery.

4 CHAPTER FOUR Clinical features

Epistaxis and obstruction in a boy.

4.1 Nasal and nasopharyngeal

- **Profuse, recurrent, spontaneous epistaxis:** the commonest presentation, often with anaemia.
- **Progressive nasal obstruction** with a hyponasal voice.
- **Nasal discharge** from obstruction and sinusitis.

4.2 Ear

Conductive hearing loss and otitis media with effusion from tubal obstruction.

4.3 Eye and nerves

Proptosis and diplopia; involvement of the second, third, fourth and sixth nerves with orbital and intracranial spread.

4.4 Face

Broadening of the nasal bridge, swelling of the cheek, and a *frog face* with bilateral proptosis.

4.5 Examination

Endoscopy shows a smooth or lobulated, pink to purplish mass with prominent surface vessels in the back of the nose and nasopharynx, obstructing one or both choanae, sometimes pushing the soft palate down or hanging into the oropharynx. Never palpate it with a finger outside the operating theatre.

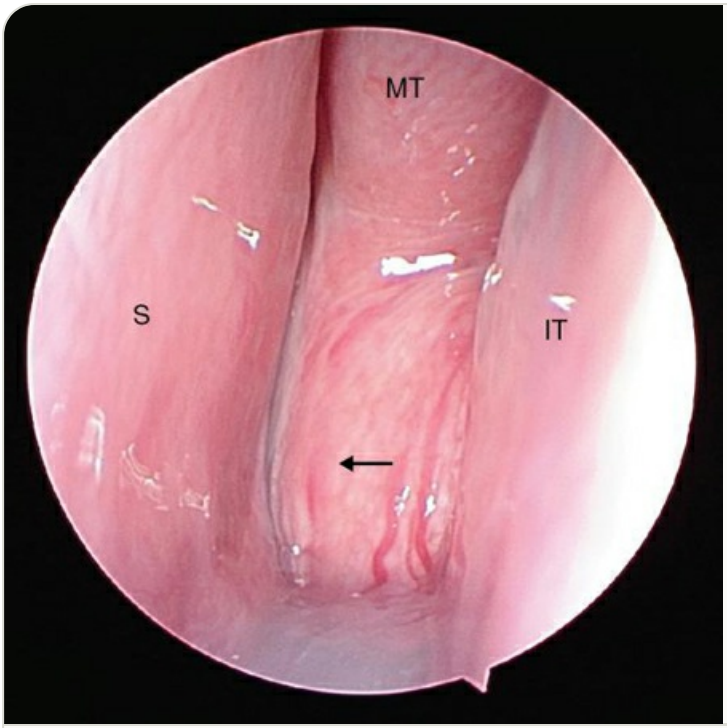


FIGURE 5 Endoscopic view of an angiofibroma (arrow) in the left nasal cavity, with prominent surface vessels; septum (S), inferior turbinate (IT), middle turbinate (MT).

Dhingra, fig. 49.5

NEVER BIOPSY IN CLINIC

Diagnosis is made by imaging. If tissue is truly needed to exclude another tumour, biopsy is done under general anaesthesia with blood cross-matched and everything ready to control bleeding.

5 CHAPTER FIVE Investigations

Imaging makes the diagnosis.

5.1 Contrast CT

The investigation of choice. It shows the enhancing mass, bone erosion and the extent of spread. *Anterior bowing of the posterior wall of the maxillary sinus* (the antral or Holman-Miller sign) is pathognomonic. Widening of the pterygopalatine fossa and erosion of the vidian canal are typical.

5.2 MRI

Complements CT for soft tissue extension: intracranial, cavernous sinus, orbit, infratemporal fossa. Flow voids give a *salt and pepper* appearance.

5.3 Angiography and embolisation

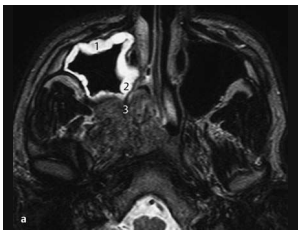
Digital subtraction angiography shows the feeders, mainly from the external carotid (the internal maxillary artery's sphenopalatine and posterior nasal branches, the ascending pharyngeal and ascending palatine arteries), and from the internal carotid in large or intracranial tumours. External carotid feeders can be embolised; surgery should follow within 24 to 48 hours, before collaterals develop. Embolisation carries risks of stroke, blindness and cranial nerve palsy, so some surgeons omit it for small tumours.



FIGURE 6 First, the specimen of an extensive angiofibroma and its CT, showing bone destruction and spread into the pterygopalatine fossa. Second, angiography before and after embolisation.
Dhingra, figs 49.6 and 49.7

IMAGING IN THE APP 1 case

Juvenile angiofibroma on CT, MRI and angiography.



Angiofibroma, Juvenile
Tumours · 12 images

5.4 Preparation

Group and cross-match two to three units of blood.

5.5 Differential diagnosis

Antrochoanal polyp, enlarged adenoids, olfactory neuroblastoma, rhabdomyosarcoma, lobular capillary haemangioma and haemangiopericytoma.

6 CHAPTER SIX Staging

Extent decides the approach.

Several systems exist (Sessions, Fisch, Chandler, Radkowski, Andrews, and newer endoscopic systems), all based on extension into the sinuses, pterygopalatine and infratemporal fossae, orbit and cranial cavity, and on residual vascularity after embolisation.

7 CHAPTER SEVEN Treatment

Surgery is the treatment.

7.1 Principles

Complete surgical removal is the treatment of choice; the tumour does not regress with age. Embolisation reduces blood loss. Hormonal treatment is no longer used.

7.2 Approaches by extent

EXTENT AND APPROACH

EXTENT	APPROACH
Nose and nasopharynx	Endoscopic, or transpalatal
Plus sinuses and pterygopalatine fossa	Endoscopic, lateral rhinotomy with medial maxillectomy, midfacial degloving, or Le Fort I
Plus infratemporal fossa	Extended endoscopic, infratemporal fossa approach, or maxillary swing
Plus cheek	Maxillary swing (facial translocation)
Plus intracranial extension	Combined endoscopic or open skull base surgery with neurosurgery; radiotherapy if inaccessible
Residual or recurrent	Observation, revision surgery, or radiotherapy

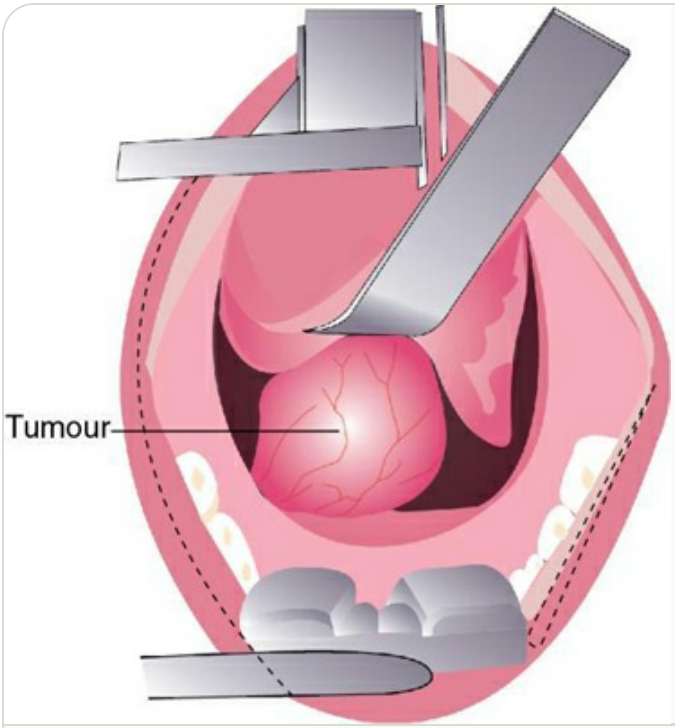


FIGURE 7 Angiofibroma exposed through a transpalatal approach.
Dhingra, fig. 49.8

7.2.1 Endoscopic surgery

Now used for most tumours, including many with infratemporal and limited intracranial extension, with less bleeding, no facial scars and fewer recurrences.

7.2.2 Open approaches

Transpalatal (with Sardana's sublabial extension), lateral rhinotomy and midfacial degloving with medial maxillectomy, the Le Fort I approach, the maxillary swing (Wei's facial translocation), the infratemporal fossa approach, and combined intracranial-extracranial resection.

OPERATIVE GUIDES IN THEATRE 2 operations

Juvenile Angiofibroma
Juvenile angiofibroma
Endoscopic Endonasal Transpterygoid Approaches to the Middle Cranial Fossa
Transpterygoid approach

7.3 Radiotherapy

For unresectable intracranial disease, residual tumour next to critical structures, and recurrence; stereotactic radiosurgery for small residuals. Long-term risks

include growth disturbance, cataract and radiation-induced malignancy in young patients.

7.4 Recurrence

Up to about a third, usually within two years, from residual tumour in the basisphenoid and vidian canal. Follow with contrast MRI.

UPMC STAGING AND CURRENT MANAGEMENT (OXFORD TEXTBOOK)

Older systems (Fisch, Radkowski, Andrews) stage angiofibroma by extent only. The **University of Pittsburgh Medical Center (UPMC)** system adds residual vascularity from the internal carotid artery after embolisation, and predicts blood loss and recurrence better. Surgery is the preferred treatment. Early tumours are removed endoscopically, often without embolisation; advanced tumours with internal carotid supply have the external carotid branches embolised first. Recurrence rates are similar after open and endoscopic surgery. Small residual tumour may be watched for growth; accessible recurrence is removed endoscopically, and disease in high-risk sites such as the cavernous sinus may be observed or treated with stereotactic radiosurgery. Follow-up with imaging continues for at least three years.

UPMC STAGING OF ANGIOFIBROMA

STAGE	EXTENT
I	Nasal cavity, medial pterygopalatine fossa
II	One or more sinuses, lateral pterygopalatine fossa; no residual vascularity
III	Skull base erosion, orbit or infratemporal fossa; no residual vascularity
IV	Skull base erosion, orbit or infratemporal fossa; residual vascularity
V	Intracranial extension with residual vascularity (medial or lateral)

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The patient:** adolescent boy, epistaxis and obstruction.
- **Site of origin** and routes of spread through the pterygopalatine fossa.
- **The antral (Holman-Miller) sign.**
- **Why no biopsy,** and the role of embolisation and its timing.
- **Surgical approaches** by extent.

8.2 Common pitfalls

COMMON PITFALLS

- Biopsying the mass in clinic.
- Treating recurrent epistaxis in a teenage boy without endoscopy.
- Leaving tumour in the vidian canal.

Written for this app from Oxford Textbook of Otolaryngology ch. 39, Oxford Specialist Handbook ch. 20, Dhingra ch. 49 and K. J. Lee ch. 32. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

NOSE & NASAL CAVITY · TUMOURS

Angiofibroma, Juvenile

IMAGING FINDINGS

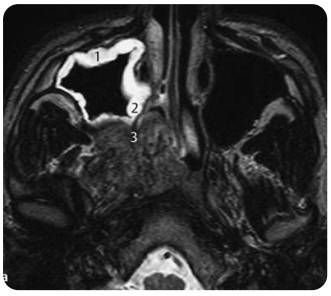
- Enhancing mass at the sphenopalatine foramen in an adolescent male
- Widened pterygopalatine fossa; bowing of the posterior maxillary wall
- Flow voids; extension to the sphenoid, infratemporal fossa and skull base

TOP DIFFERENTIALS

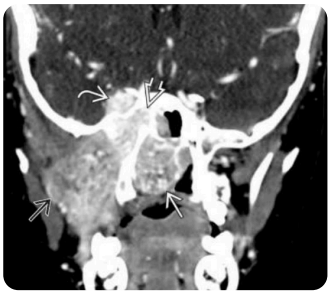
- Hypervascular polyp
- Rhabdomyosarcoma
- Antrochoanal polyp

CHECKLIST

- No biopsy: angiography and embolisation before surgery
- Vidian canal and cavernous sinus spread



A young man with obstruction and bloodtinged nasal discharge from the right side of the nose.
Dunnebier, Imaging for…



Coronal CECT in a 13-year-old shows typical growth pattern of a large, hypervascular angiofibroma occluding the nasopharynx, extending into righ…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 740.